RCF V2.1 · CONSOLIDATED GAP AUDIT · ALL 8 SECTIONS

Gap Audit & Phase C

Consolidated audit of all 31 open gaps surfaced during the v2.1 section-by-section rewrite (§0–§7). Severity-ranked: 7 FATAL (red [illegible] structural issues), 23 MATERIAL, 1 LOW. Includes dependency-chain analysis (T-2 resolves 39% of all gaps; D=3 resolves 16%; Friedmann resolves 10%), gap clustering, what v2.1 introduced vs. pre-existing, and the recommended Phase C attack order. The framework's open problems are not 31 scattered gaps — they cluster into 3 fundamental questions.

31 GAPS · 3 FATAL ISSUES

31	3	39%	8
TOTAL GAPS	FATAL ISSUES	RESOLVED BY T-2	SECTIONS AUDITED

VERSION	v2.1 Draft (Canonical Form)
AUDIENCE	Internal Framework Team
DATE	2026-08-02

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RCF v2.1 Consolidated Gap Audit

All Open Gaps Surfaced During the 8-Section Canonical-Form Rewrite

Preamble

This document consolidates every open gap surfaced during the v2.1 section-by-section rewrite of the Reconciliation Causal Framework (§§0–7). Each gap is listed with its ID, location, description, severity, what it blocks, and its dependency chain. The gaps are severity-ranked: FATAL first, then MATERIAL, then LOW.

The headline finding: **31 total gaps, of which 7 are FATAL — but the 7 fatal gaps reduce to 3 distinct structural issues.** The framework's open problems are not 31 scattered gaps; they cluster into 3 fundamental questions that Phase C must address.

The 3 Distinct Fatal Issues

The 7 FATAL gaps cluster into 3 distinct structural questions. These are the only issues that can falsify the framework's central claims if not resolved.

Fatal Issue 1: D=3 Representation Theory

The question: Does the Jordan algebra `A_J` support exactly 3 independent symmetric bilinear invariants?

Why it matters: The D=3 closure (Theorem 3.6.4) depends on this representation-theoretic claim. If `A_J` supports more than 3 independent invariants, the framework predicts extra spatial dimensions or a modified geometry — a major structural problem. If it supports fewer, the geometry is under-determined.

Gaps in this cluster:

Gap ID	Location	Description
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GAP-0.1.5	§0.1 Remark 0.1.5	First flagged at the Lie–Jordan decomposition — the algebraic foundation
GAP-3.6.5	§3.6 Remark 3.6.5	Primary location — the D=3 closure argument itself
GAP-5.3.3	§5.3 Remark 5.3.4	Propagated — the holographic boundary depends on D=3

Dependency chain:

- Lie–Jordan decomposition (§0.1.2) → Jordan algebra `A_J` → representation theory question (GAP-0.1.5/3.6.5) → D=3 closure (Thm 3.6.4) → dimensional suppression (Thm 5.3.1) → holographic boundary (Thm 5.3.3) → entropy-area scaling (Thm 5.5.4)

What blocks if unresolved: D=3 closure, holographic boundary derivation, BH entropy mechanism (area scaling via channel suppression), the entire §5 black-hole framework.

Resolution path: Develop the representation theory of Jordan algebras over $\mathbb{C}$ (or $\mathbb{R}$) for the framework's possibly-infinite-dimensional A. The Jordan–von Neumann–Wigner classification (1934) gives the finite-dimensional structure theory; the infinite-dimensional extension needs development. This is a HARD mathematical problem — possibly the hardest in the framework.

Fatal Issue 2: T-2 Spectral Gap + Adjacency Structure

The question: Does `F̂` have a spectral gap on `ker(M̂)` in the infinite-dimensional case, and does the constraint adjacency graph (from Jordan co-occurrence) have bounded degree?

Why it matters: T-2 is the **existential open problem** — the single point of failure for the majority of the framework's physical predictions. If T-2 fails, the Convergence Theorem, thin=full, mass-burden identity, Dirac bracket, `Λ_B = 0`, Einstein closure, Friedmann equation, and BH holography all collapse from Established/Conditional to open.

Gaps in this cluster:

Gap ID	Location	Description
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GAP-0.1.7	§0.1 Remark 0.1.7	Adjacency structure on constraint index set I not yet defined
GAP-1.1.7	§1.1 Remark 1.1.7	Adjacency graph G_I bounded degree not proven
GAP-1.6.1	§1.6 Remark 1.6.2	T-2 three prerequisites: local gap, uniformity, bounded degree — all open

Dependency chain:

- Jordan co-occurrence (`Ĉ_α ∘ Ĉ_β ≠ 0`) → adjacency graph `G_I` (Def 1.1.6) → bounded degree (GAP-1.1.7) → Lieb-Robinson bound (Thm 1.1.4, T-1) + detectability lemma (Thm 1.6.1, T-2) → Convergence Theorem → Master-Zero → `Λ_B = 0` → Einstein closure → Friedmann → everything downstream

What blocks if unresolved: T-1 (γ derivation, Lieb-Robinson), T-2 (spectral gap, Convergence Theorem), and every T-2-conditional result: thin=full, mass-burden, Dirac bracket, `Λ_B = 0`, Einstein closure, Friedmann, BH holography, Arrow of Time, Born rule (T-4 STRENGTHENED conditional on T-2).

Resolution path:

1. **Prove bounded degree directly from Jordan algebra structure** — without assuming D=3. This requires showing the Jordan co-occurrence graph is locally finite. Plausible because each constraint interacts with finitely many others (the local algebra per cell is `M_d(ℂ)`), but needs proof.

2. **Verify local gap** `γ_local > 0` on the specific constraint algebra of RCF (not just the `M_6(ℂ)` toy model).

3. **Prove uniformity** — γ_{local} doesn't shrink under Open Expansion (the local algebra is always $M_d(\mathbb{C})$, so the local gap is always the same).

4. **Apply the detectability lemma** (Aharonov–Arad–Landau–Vazirani 2009) to lift local gaps to global: $\gamma_{global} \geq \gamma_{local}/(C \cdot \Delta)$.

This is the highest-priority Phase C task. Closing T-2 converts the majority of conditional results to Established.

Fatal Issue 3: Reconciliation Principle Derivation

The question: Can the Reconciliation Principle (Principle 2.1.1) be derived from the Open Expansion Principle (Postulate 1.1.1), or is it an irreducible new primitive?

Why it matters: If RP is irreducible, the framework has more primitives than its own guardrails permit (Guardrail 4: "No new primitives without explicit justification"). If RP is derived, the derivation is currently missing. Either way, the framework's primitive count is not honest until this is resolved.

Gap:

Gap ID	Location	Description
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GAP-2.1.2	§2.1 Remark 2.1.2	RP derivation from Open Expansion missing; violates Guardrail 4 if irreducible

Dependency chain:

- Open Expansion Principle (Post 1.1.1) → [missing derivation] → Reconciliation Principle (Princ 2.1.1) → variational target $I(S)$ → RP well-posedness (Thm 2.1.6, T-2-conditional) → fields, particles, mass, records, Born rule, FIREWALL

What blocks if unresolved: The framework's primitive count honesty. If RP is irreducible, it must be explicitly listed alongside Postulate 1.1.1 in Guardrail 4. If RP is derived, the derivation must show that $I(S) = \Sigma [s(a,b) - s^*(a,b)]^2$ is the unique functional whose minimization enforces Lie–Jordan compatibility under finite-rate Open Expansion.

Resolution path:

- **Option (a) — DERIVE RP from Open Expansion** (preferred): show that minimizing $I(S)$ is equivalent to enforcing Lie–Jordan compatibility. This would make RP a theorem, not a primitive.

- **Option (b) — EXPLICITLY LIST RP as a primitive** (honest fallback): update Guardrail 4 to include RP alongside the other 8 primitives. This is honest but weakens the framework's minimality claim.

All 31 Gaps (Severity-Ranked)

FATAL (7 gaps, 3 distinct issues)

#	Gap ID	Section	Description	Blocks	Distinct Issue
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1	GAP-0.1.5	§0.1	Representation-theoretic claim "A_J supports exactly 3 invariants" not proven	D=3 closure	Fatal Issue 1
2	GAP-3.6.5	§3.6	Same as GAP-0.1.5 at primary location (D=3 closure argument)	D=3 closure, BH entropy	Fatal Issue 1
3	GAP-5.3.3	§5.3	Dimensional suppression inherits D=3 dependency	Holographic boundary	Fatal Issue 1
4	GAP-0.1.7	§0.1	Adjacency structure on constraint index set I not yet defined	T-1, T-2	Fatal Issue 2
5	GAP-1.1.7	§1.1	Adjacency graph G_I bounded degree not proven	T-1 (Lieb-Robinson), T-2 (detectability)	Fatal Issue 2
6	GAP-1.6.1	§1.6	T-2 three prerequisites: local gap, uniformity, bounded degree — all open	Convergence Theorem + all T-2-conditional results	Fatal Issue 2
7	GAP-2.1.2	§2.1	RP derivation from Open Expansion missing (violates Guardrail 4 if irreducible)	Primitive count honesty	Fatal Issue 3

MATERIAL (23 gaps)

#	Gap ID	Section	Description	Blocks
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8	GAP-0.2.7	§0.2	Jordan component Ĉ_α ∘ Ĉ_β named but not given operational meaning	Adjacency structure (§1.1)
9	GAP-0.3.4	§0.3	K_ω vs K_ω^J: which is the correct kernel?	§3.1, D=3 (§3.6)

#	Gap ID	Section	Description	Blocks
10	GAP-0.3.7	§0.3	F decomposition named but not formalized	Relational burden (§4.4)
11	GAP-1.2.4	§1.2	Fracture existence depends on T-2	F decomposition, relational burden
12	GAP-1.4.7	§1.4	Dirac bracket RCF-specific verification depends on fracture	Dirac bracket (§1.4.6)
13	GAP-1.5.4	§1.5	Lindblad form T-2-conditional; not propagated to downstream	Arrow of Time (§4.1.4), decoherence (§2.6.3)
14	GAP-2.1.6	§2.1	RP well-posedness T-2-conditional	RP rigor (propagated)
15	GAP-2.2.7	§2.2	Dark energy cosmological predictions conditional on Friedmann (§6.2.2)	Observational comparison (§6)
16	GAP-2.3.4	§2.3	Anti-hermiticity of burden flux T-2-conditional	Covariant Laplacian self-adjointness
17	GAP-2.3.7	§2.3	Wave equation is candidate, not derived	Field dynamics (T-2/T-3)
18	GAP-2.7.5	§2.7	Born rule full proof (Petz/Bures direction) incomplete	T-4 closure
19	GAP-3.1.4	§3.1	K_ω vs K_ω^J coexistence (Born rule vs spatial geometry) not resolved	Unity of geometric picture
20	GAP-3.1.6	§3.1	Real (K_ω^J) vs complex (K_ω) Gram determinant sufficiency	D=3 closure technical details
21	GAP-3.6.7	§3.6	K_ω^J real Gram determinant sufficiency for D=3	D=3 closure proof
22	GAP-3.8.1	§3.8	Smooth field convergence T-2/T-3-conditional	Continuum limit (propagated)
23	GAP-4.1.4	§4.1	Arrow of Time T-2-conditional (Lindblad form)	Arrow of Time (propagated)

#	Gap ID	Section	Description	Blocks
24	GAP-4.4.2	§4.4	Coincidence problem: structural argument only, quantitative match needs simulation	DM/DE/baryon density match (Q-1)
25	GAP-4.4.4	§4.4	(Sector count)^2 scaling: K_ω decay rate needs verification	DM halo profile (Q-1)
26	GAP-4.6.3	§4.6	Lorentz factor: leading order derived, full form needs exact calculation	Full Lorentz group (T-1)
27	GAP-5.5.4	§5.5	BH coefficient 1/4: existence-channel binary assumption needs verification (T-6)	T-6 closure
28	GAP-6.2.2	§6.2	Friedmann derivation path stated, not executed (M5)	Q-13 ΛCDM recovery
29	GAP-6.3.2	§6.3	Dark energy cosmological predictions conditional on Friedmann (M2)	Observational comparison
30	GAP-7.4.1	§7.4	λ_R is a new free parameter, not derived	S_eff minimality

LOW (1 gap)

#	Gap ID	Section	Description	Blocks
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31	GAP-1.1.3	§1.1	δJ formal decomposition (J_target vs J_current) not rigorous	Cosmetic (δJ = −B_Δ is consistent)

Gap Distribution by Section

Section	Fatal	Material	Low	Total
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§0 (Algebraic Foundation)	2	3	0	5
§1 (Open Expansion, Fracture, R_t)	2	3	1	6

Section	Fatal	Material	Low	Total
§2 (Burden, Matter, RP)	1	5	0	6
§3 (Emergent Geometry)	1	4	0	5
§4 (Time, Gravity, Three-Tier Burden)	0	4	0	4
§5 (Black Holes)	1	1	0	2
§6 (Cosmology)	0	2	0	2
§7 (Audit, Closure)	0	1	0	1
Total	**7**	**23**	**1**	**31**

Gap Clustering Analysis

The 31 gaps are not independent — they cluster into dependency chains. Here are the main clusters:

Cluster A: T-2 Dependency Chain (12 gaps)

Everything conditional on T-2. If T-2 closes, these 12 gaps resolve in one stroke:

GAP-0.1.7 → GAP-1.1.7 → GAP-1.6.1 (T-2 itself) → GAP-1.2.4 (fracture) → GAP-1.4.7 (Dirac) → GAP-1.5.4 (Lindblad) → GAP-2.1.6 (RP well-posedness) → GAP-2.3.4 (anti-hermiticity) → GAP-2.3.7 (wave equation) → GAP-2.7.5 (Born rule, T-4) → GAP-3.8.1 (smooth field convergence) → GAP-4.1.4 (Arrow of Time)

This is why T-2 is the EXISTENTIAL priority. Closing it resolves 12 of 31 gaps (39%).

Cluster B: D=3 Dependency Chain (5 gaps)

Everything conditional on the D=3 representation-theoretic claim:

GAP-0.1.5 → GAP-3.6.5 (D=3 itself) → GAP-5.3.3 (holographic boundary) → GAP-3.1.6 (real vs complex Gram) → GAP-3.6.7 (K_ω^J Gram sufficiency)

Closing D=3 resolves 5 of 31 gaps (16%).

Cluster C: Friedmann Dependency Chain (3 gaps)

Everything conditional on the Friedmann derivation:

GAP-6.2.2 (Friedmann itself) → GAP-2.2.7 (dark energy cosmological) → GAP-6.3.2 (dark energy predictions)

Executing the Friedmann derivation resolves 3 of 31 gaps (10%).

Cluster D: K_ω^J Adoption (2 new gaps)

Introduced by the v2.1 Fubini-Study swap:

GAP-3.1.4 (K_ω vs K_ω^J coexistence) → GAP-3.1.6 (real vs complex Gram)

These are NEW gaps from v2.1 — both MATERIAL, neither FATAL. The Fubini-Study swap itself introduced NO fatal gaps (it eliminated the Factorization + Dominance apparatus).

Cluster E: Simulation-Dependent (3 gaps)

Require numerical simulation, not analytical derivation:

GAP-4.4.2 (coincidence problem quantitative match) → GAP-4.4.4 (K_ω decay rate / DM halo) → GAP-4.6.3 (Lorentz factor full form)

Standalone Gaps (7 gaps)

Not part of a chain — independent issues:

GAP-0.2.7 (Jordan component operational meaning), GAP-0.3.4 (K_ω vs K_ω^J choice), GAP-0.3.7 (F decomposition — resolved by §1.2.4), GAP-2.1.2 (RP derivation — Fatal Issue 3), GAP-5.5.4 (BH 1/4 binary assumption), GAP-7.4.1 (λ_R free parameter), GAP-1.1.3 (δJ notation — LOW)

What v2.1 Introduced vs. What Was Pre-Existing

NEW gaps introduced by the v2.1 rewrite (2 gaps, both MATERIAL):

Gap ID	Description	Why introduced
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GAP-3.1.4	K_ω vs K_ω^J coexistence	The K_ω^J Jordan-built kernel adoption (F2 fix) raised the question of whether it's the correct kernel for ALL downstream geometry or only for spatial geometry
GAP-3.1.6	Real vs complex Gram determinant	K_ω^J is real-valued; the Gram determinant's linear-dependence detection may need the complex phase structure of $\tilde{K}_\omega$

Assessment: ***The v2.1 rewrite introduced 2 new MATERIAL gaps (neither FATAL) while eliminating the Factorization + Dominance apparatus (which was a source of v2.0 complexity and a conditional-metricity burden). Net effect:*** *the rewrite reduced structural complexity while introducing 2 manageable technical questions.*

Pre-existing gaps from the patch plan / Phase C program (29 gaps):

All other 29 gaps were pre-existing — identified in the v2.0→v2.1 patch plan (15 issues) or the Phase C mathematical program (6 targets). The v2.1 rewrite has now flagged them at their point of origin in the canonical text.

Recommended Phase C Attack Order

Based on the clustering analysis:

Priority 1: T-2 (resolves 12 gaps — 39% of all open issues)

Attack path:

1. Prove adjacency graph bounded degree from Jordan algebra structure (GAP-1.1.7).

2. Verify local gap γ_local > 0 on the specific RCF constraint algebra (GAP-1.6.1).

3. Prove uniformity of γ_local under Open Expansion (GAP-1.6.1).

4. Apply the detectability lemma to get global gap (GAP-1.6.1 → T-2 CLOSED).

Impact: Converts Convergence Theorem, thin=full, mass-burden, Dirac bracket, Λ_B=0, Einstein closure, Arrow of Time, Born rule (T-4), and 4 other results from Conditional to Established.

Priority 2: D=3 (resolves 5 gaps — 16% of all open issues)

Attack path:

1. Develop the representation theory of Jordan algebras for the framework's A (GAP-3.6.5).

2. Prove A_J supports exactly 3 independent symmetric bilinear invariants.

3. Verify the Gram-rank argument with K_ω^J (GAP-3.1.6, GAP-3.6.7).

Impact: Converts D=3 closure from Conditional Proposition to Theorem; secures the holographic boundary and BH entropy mechanism.

Priority 3: Friedmann execution (resolves 3 gaps — 10% of all open issues)

Attack path:

1. Execute the 4-step FLRW ADM variation (GAP-6.2.2).

2. Compute the homogeneous/isotropic burden tensor components.

3. Vary S_eff with respect to a(t).

4. Calibrate $8\pi G/3$ via κ_B.

Impact: Upgrades Friedmann from Conjecture to Conditional Theorem; converts dark energy cosmological predictions from CONDITIONAL to Established (conditional on T-2).

Priority 4: RP derivation (resolves 1 FATAL gap — Fatal Issue 3)

Attack path:

- Either derive RP from Open Expansion (preferred) or explicitly list RP as a primitive (fallback).

Impact: Resolves the primitive-count honesty question; removes the F3 flag.

Priority 5: K_ω^J resolution (resolves 2 NEW gaps)

Attack path:

- Determine whether K_ω^J (Jordan-built) is the correct kernel for ALL downstream geometry, or whether K_ω (sesquilinear) is needed for some applications (e.g., the Born rule's projective geometry).

Impact: Resolves the 2 new gaps introduced by the v2.1 Fubini-Study swap; unifies the geometric picture.

Lower priority: Simulation-dependent gaps (3 gaps)

GAP-4.4.2, GAP-4.4.4, GAP-4.6.3 require numerical simulation of Open Extension descent on FLRW — a computational physics project, not an analytical derivation.

Summary Assessment

The v2.1 rewrite has:

1. **Applied all patches** from the v2.0→v2.1 patch plan (9 mandatory + 3 hard).

2. **Applied the Phase C mathematical program** at the strategy level (Fubini-Study metric, K_ω^J Jordan-built kernel, 𝔽 decomposition, γ=√λ_min(𝔽), adjacency structure, existence-channel suppression, Petz/Bures Born rule, Lorentz factor from burden, coincidence problem, channel-suppression entropy).

3. **Surfaced 31 gaps** at their point of origin in the canonical text.

4. **Reduced the 7 FATAL gaps to 3 distinct structural issues** — the framework's open problems are not scattered; they cluster.

5. **Introduced only 2 NEW gaps** (both MATERIAL, neither FATAL) — the Fubini-Study swap's K_ω^J adoption raised technical questions but did not create structural problems.

6. **Verified acyclicity** across all 7 sections (Theorem 7.5.1).

The v2.1 manuscript is architecturally complete and internally consistent within its stated hypotheses. The remaining work is targeted mathematical and phenomenological — 3 distinct fatal issues (T-2, D=3, RP derivation), plus the Friedmann execution and the K_ω^J resolution. If T-2 closes, 39% of all open gaps resolve in one stroke.

The foundational claim stands:

> A physical structure is an admissible relational structure whose constraint violations vanish in the physical state. `ω(Mˆ) = 0`.

Phase C begins.

End of Consolidated Gap Audit.